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## 1. Problem Description

Access to timely healthcare support remains an important community challenge. People may experience health problems but may not always know how urgent their situation is or what level of healthcare attention they should seek.

In some situations, delays in seeking appropriate healthcare can make a health problem more difficult to manage. A simple logical system can therefore help organize basic information about a person's reported symptoms and identify the urgency of the situation.

The proposed **Healthcare Sierra Leone** system is designed as a basic healthcare reporting and triage-support system. The system collects information such as the person's age, symptoms, temperature, and severity of symptoms. It processes this information using decision structures and classifies the situation into different urgency levels.

The system is **not intended to diagnose diseases**. Instead, it provides a logical indication of urgency and recommends that the user seek an appropriate level of healthcare attention.

## 2. Background

Healthcare services are important to the wellbeing of individuals and communities. When people experience symptoms, they need to understand when they should monitor the situation and when they should seek professional medical attention.

A logical computer-based system can help demonstrate how information can be collected, validated, processed, and used to produce a clear output.

HealthCare Sierra Leone demonstrates this concept through programming logic. The system accepts basic information from a user, validates the information, evaluates the reported conditions, and produces an urgency level.

The project focuses on **logical problem solving rather than medical diagnosis**.

## 3. Community Impact

- Healthcare challenges can affect many people within a community, including:
- Children and young people.
- Adults and elderly people.
- Families and caregivers.
- Students.
- Workers.

People living in communities with limited access to healthcare services.

A person who does not recognize the urgency of a health situation may delay seeking professional help.

HealthCare Sierra Leone can help demonstrate a simple way of organizing basic health information and communicating an appropriate urgency recommendation.

The system can also serve as an educational example of how computational thinking can be applied to a real-world healthcare problem.

## 4. Proposed Solution

The proposed solution is **HealthCare Sierra Leone**, a Community Healthcare Reporting and Triage Support System.

The system collects basic information from the user:

- Age
- Main symptom
- Temperature
- Symptom severity

The information is then validated.

If invalid information is entered, the system displays:

**Invalid input. Please try again.**

If the information is valid, the system processes the reported conditions.

The system classifies the situation as:

- LOW
- MODERATE
- HIGH
- EMERGENCY

The system then provides an appropriate recommendation.

For example:

- **LOW:** Monitor the situation and consider routine healthcare advice.
- **MODERATE:** Contact or visit a healthcare professional.
- **HIGH:** Seek medical attention as soon as possible.
- **EMERGENCY:** Seek emergency medical assistance immediately.

The user can also choose whether to submit another report.

## 5. COMPUTATIONAL THINKING PROCESS

### Decomposition

- The healthcare problem is divided into smaller tasks:
- Collect patient information.
- Validate the information.
- Analyze the reported symptoms.
- Determine urgency.
- Generate a recommendation.
- Display the result.
- Allow another report.
- End the system.

**Pattern Recognition**

The system looks for patterns in the information provided.

For example, a combination of **severe symptoms and a high temperature** may indicate that the situation requires a higher level of attention.

The system uses these conditions to determine the appropriate urgency category.

**Abstraction**

The system only collects information that is necessary for the logical process:

- Age
- Main symptom
- Temperature
- Symptom severity

Unnecessary personal information is not included.

**Algorithmic Thinking**

The system follows a clear sequence:

Input → Validate → Analyze → Determine Urgency → Generate Recommendation → Display Result → Repeat or End

## 6. ALGORITHM DESIGN

- The algorithm begins by collecting basic healthcare information from the user.
- The information is then validated to make sure that the required values are acceptable.
- If invalid information is entered, the system displays an error message and asks the user to enter the information again.
- After successful validation, the system evaluates the reported conditions using decision structures.
- The system assigns one of four urgency levels:

**LOW → MODERATE → HIGH → EMERGENCY**

A recommendation is then generated according to the urgency level.

The system displays the result and asks whether another report should be submitted.

## 7. SDG ALIGNMENT

**Sustainable Development Goal 3 -** Good Health and Well-Being

HealthCare Sierra Leone directly relates to **SDG 3: Good Health and Well-Being**.

The project promotes awareness of health conditions and encourages people to seek appropriate professional healthcare attention when necessary.

The system also demonstrates how technology and logical problem-solving can be applied to support healthcare awareness.

## 8. OPEN SOURCE JUSTIFICATION

Health SierraCare Leone could be developed as an Open Source project so that students, developers, educational institutions, and community organizations can study and improve the system.

**Open Source** development could provide benefits such as:

- Collaboration.
- Educational use.
- Improvement of the system.
- Sharing of ideas.
- Adaptation to different communities.

The project's algorithm, pseudocode, flowchart, and documentation could be made available through a GitHub repository.

## 9. DIGITAL PUBLIC JUSTIFICATION

HealthCare Sierra Leone could potentially be developed as a Digital Public Good if it follows appropriate principles of accessibility, openness, responsible use, documentation, and privacy.

The system could be made available for educational institutions and community organizations to study and adapt.

- Its documentation could include:
- Algorithm.
- Pseudocode.
- Flowchart.
- Functional modules.
- User instructions.
- Project documentation.

The system would remain a support tool rather than a replacement for qualified healthcare professionals.

## 10. ALGORITHM

**HealthCare Sierra Leone Algorithm**

**Step 1:**Start the system.

**Step 2:** Display the HealthCare Sierra Leone welcome message.

**Step 3:** Enter the person's age.

**Step 4:**Enter the main symptom.

**Step 5:**Enter the temperature.

**Step 6:**Enter the symptom severity.

**Step 7:** Validate the information.

**Step 8:** If the information is invalid, display:

**Invalid input. Please try again.**

**Step 9:**Return to the input stage.

**Step 10:**If the information is valid, analyze the reported health conditions.

**Step 11:** If the situation indicates an emergency, set:

`urgencyLevel = "EMERGENCY"`

**Step 12:** Otherwise, if the situation indicates high urgency, set:

`urgencyLevel = "HIGH"`

**Step 13:**Otherwise, if the situation indicates moderate urgency, set:

`urgencyLevel = "MODERATE"`

**Step 14:**Otherwise, set:

`urgencyLevel = "LOW"`

**Step 15:**Generate an appropriate recommendation.

**Step 16:** Display the age, symptom, urgency level, and recommendation.

**Step 17:** Ask whether the user wants to submit another report.

**Step 18:** If yes, return to the input stage.

**Step 19:** If no, display the thank-you message.

**Step 20:**End the system.

## 11. PSEUDOCODE

```
BEGIN
    DECLARE age AS INTEGER
    DECLARE symptom AS STRING
    DECLARE temperature AS REAL
    DECLARE severity AS STRING
    DECLARE urgencyLevel AS STRING
    DECLARE recommendation AS STRING
    DECLARE isValid AS BOOLEAN
    DECLARE submitAnotherReport AS STRING

    SET submitAnotherReport TO "yes"
    WHILE submitAnotherReport = "yes"
        DISPLAY "Welcome to HealthCare Sierra Leone"
        SET isValid TO FALSE
        WHILE isValid = FALSE
            DISPLAY "Enter age:"
            INPUT age
            DISPLAY "Enter main symptom:"
            INPUT symptom
            DISPLAY "Enter temperature:"
            INPUT temperature
            DISPLAY "Enter symptom severity:"
            INPUT severity

            SET isValid TO TRUE

            IF age <= 0 THEN
                SET isValid TO FALSE
            END IF

            IF symptom IS EMPTY THEN
```

```
        SET isValid TO FALSE
    END IF

    IF temperature <= 0 THEN
        SET isValid TO FALSE
    END IF
    IF severity IS NOT "Mild",
       "Moderate", OR "Severe" THEN
        SET isValid TO FALSE
    END IF

    IF isValid = FALSE THEN
        DISPLAY "Invalid input. Please try again."
    END IF

END WHILE

IF severity = "Severe" AND temperature >= 39.0 THEN

    SET urgencyLevel TO "EMERGENCY"

ELSE IF severity = "Severe" THEN

    SET urgencyLevel TO "HIGH"

ELSE IF severity = "Moderate" OR temperature >= 38.0 THEN

    SET urgencyLevel TO "MODERATE"

ELSE

    SET urgencyLevel TO "LOW"
```

```
        END IF

        IF urgencyLevel = "EMERGENCY" THEN

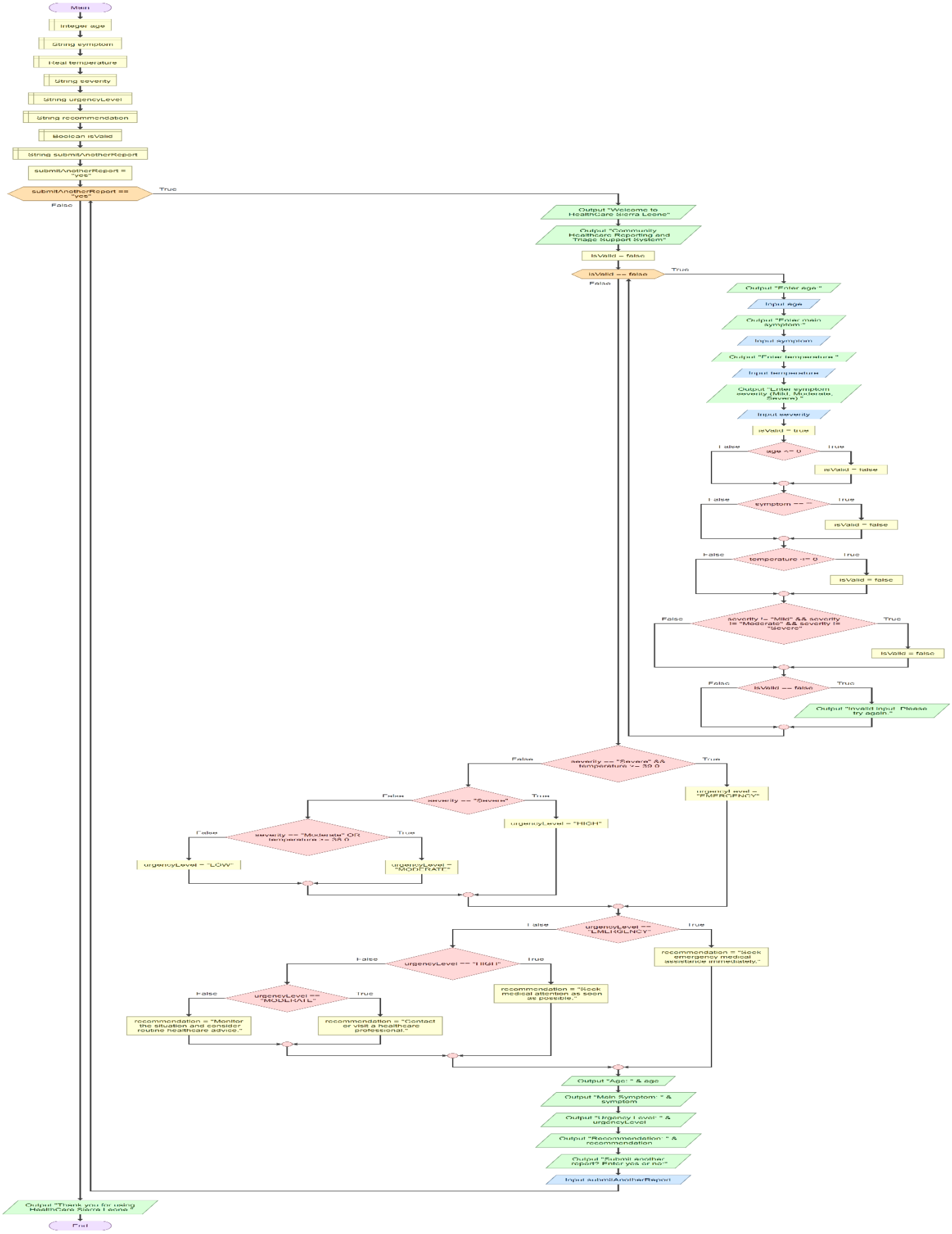
            SET recommendation TO
            "Seek emergency medical assistance immediately."

        ELSE IF urgencyLevel = "HIGH" THEN

            SET recommendation TO
            "Seek medical attention as soon as possible."
        ELSE IF urgencyLevel = "MODERATE" THEN
            SET recommendation TO
            "Contact or visit a healthcare professional."
        ELSE
            SET recommendation TO
            "Monitor the situation and consider routine healthcare advice."
        END IF
        DISPLAY "Age: " + age
        DISPLAY "Main Symptom: " + symptom
        DISPLAY "Urgency Level: " + urgencyLevel
        DISPLAY "Recommendation: " + recommendation

        DISPLAY "Submit another report? Enter yes or no:"
        INPUT submitAnotherReport
    END WHILE
    DISPLAY "Thank you for using HealthAlert Sierra Leone."
END
```

## 12. FLOWCHARTS



## 13. FUNCTIONAL MODULES

**Module 1:** Input Module

**Purpose:** Collect the basic healthcare information.

**Inputs:**

- Age
- Main symptom
- Temperature
- Symptom severity

**Module 2:** Validation Module

**Purpose:** Check whether the information entered is valid.

If invalid:

"Invalid input. Please try again."

**Module 3:** Health Condition Analysis Module

**Purpose:** Analyze the information provided by the user.

**Module 4:** Urgency Classification Module

**Purpose:** Classify the reported situation as:

- LOW
- MODERATE
- HIGH
- EMERGENCY

**Module 5:** Recommendation Module

**Purpose:** Generate an appropriate recommendation according to the urgency level.

**Module 6:** Output Module

**Purpose:** Display:

- Age
- Main symptom
- Urgency level
- Recommendation

**Module 7:** Repetition Module

**Purpose:** Allow the user to submit another report.

---

## 14. EXPECTED OUTPUTS

**Scenario 1 -** LOW

**Input**

```
```
```
```

Age: 25

Main Symptom: Headache

Temperature: 36.8

Severity: Mild

```

**Expected Output**

```
```
```
```

Age: 25

Main Symptom: Headache

Urgency Level: LOW

Recommendation:

Monitor the situation and consider routine healthcare advice.

```

**---**

**Scenario 2 -** MODERATE

**Input**

```
```
```
```

Age: 30

Main Symptom: Fever

Temperature: 38.2

Severity: Moderate

```

**Expected Output**

```

```

```

Age: 30

Main Symptom: Fever

Urgency Level: MODERATE

Recommendation:

Contact or visit a healthcare professional.

```

**---**

**Scenario 3 -** HIGH

**Input**

```

```

```

Age: 40

Main Symptom: Severe Pain

Temperature: 37.5

Severity: Severe

```

**Expected Output**

```

```

```

Age: 40

Main Symptom: Severe Pain

Urgency Level: HIGH

Recommendation:

Seek medical attention as soon as possible.

```

---

**Scenario 4 -** EMERGENCY

**Input**

```

```

```

Age: 50

Main Symptom: Severe Fever

Temperature: 39.5

Severity: Severe

```

**Expected Output**

```

```

```

Age: 50

Main Symptom: Severe Fever

Urgency Level: EMERGENCY

Recommendation:

Seek emergency medical assistance immediately.

```

---

**Scenario 5 -** INVALID INPUT

**Input**

```

```

```

Age: -5

Main Symptom:

Temperature: 38.0

Severity: Moderate

```

**Expected Output**

```

```

```

```

Invalid input. Please try again.

```

The system returns to the input stage.

## 15. REFERENCE


- World Health Organization (WHO). (2024). *Primary health care*. World Health Organization.
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